



Hepatic and metabolic outcomes induced by sub-chronic exposure to polystyrene microplastics in mice

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Abstract

Microplastics (MPs) have attracted significant attention due to their global distribution in living environments. Although some studies have reported MP-induced hepatotoxicity in mouse models, a systematic approach to MP-mediated liver toxicity was still lacking. Therefore, we used a mouse model to study the sub-chronic effects of MP exposure on the liver. Female C57BL/6 mice, aged 6 weeks, received an oral administration of 0.3 mg of Nile Red-labeled polystyrene (PS) microplastics, with particle sizes of 0.5 μm (submicron) and 5 μm (micron), via gavage, while control mice received vehicle only. Each mouse was exposed to MPs twice a week for 12 weeks. After sacrifice, the levels of MP accumulation, oxidative stress, inflammation, and pathological changes were measured in the mouse liver, and blood samples were collected for serum biochemistry analysis. Our results demonstrated that 0.5 μm PS-MPs were accumulated in mouse livers post-MP exposure, but not in the 5 μm MP exposure group. Simultaneously, increased levels of glucose, triglyceride, alanine transaminase (ALT), aspartate transaminase (AST), superoxide dismutase, 4-hydroxy-2-nonenal-mercapturic acid (HNE-MA), interleukin-6, and lipid droplets were found in the 0.5 μm MP exposure group, while the fewer responses, including elevated liver weight index, glucose, high-density lipoprotein, AST, and decreased HNE-MA were observed in 5 μm MP exposure group. These results indicate that sub-chronic exposure to submicron MPs causes MP deposition in mouse livers, which further induces oxidative stress, increases inflammatory cytokines and perturbs glucose and lipid homeostasis, which might trigger more severe metabolic dysfunction or non-alcoholic steatohepatitis-like hepatotoxicity.

Keywords Microplastics · Sub-chronic exposure · Hepatotoxicity · Mouse model · Metabolic dysfunction · Non-alcoholic steatohepatitis

Introduction

Microplastics (MPs) derived from plastic waste and products are a significant threat to human health. Numerous environmental studies have demonstrated that MPs are extensively distributed in the hydrosphere (Hurley et al. 2018; Kelly et al. 2020; Lebreton et al. 2018; Pohl et al. 2020), the atmosphere (Allen et al. 2019; Evangeliou et al. 2020), and terrestrial environments (Horton et al. 2017; Qi et al. 2020). Recently, MPs were also detected in our food and drink, including seafood (Akoueson et al. 2020; Avio et al. 2020), packed poultry meat (Kedzierski et al. 2020), vegetables (Oliveri Conti et al. 2020), seaweeds (Li et al. 2020b), salt (Kim et al. 2018), bottled water (Schymanski et al. 2018), wine (Prata et al. 2020), tea, and beer (Shruti et al. 2020). Additionally, several studies have demonstrated that food contact materials, including teabags (Hernandez et al. 2019), infant feeding bottles (Li et al. 2020a),

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take-out food containers (Du et al. 2020), and disposable paper cups (Ranjan et al. 2021), release MPs under hot conditions. As these observations indicate that our daily lives are surrounded by numerous MPs, the toxic potency of MPs on humans and other organisms should be studied.

Numerous studies have used rodent models to investigate MP-induced toxicity. MPs can enter organisms through oral intake, inhalation, and dermal contact and then accumulate inside rodent bodies (Deng et al. 2017; Fournier et al. 2020; Revel et al. 2018). The micron-, submicron-, and nano-sized MPs have been observed in diverse organs, including the intestine, liver, lung, kidney, heart, stomach, spleen, brain, testis, and ovaries (Ding et al. 2021; Walczak et al. 2015; Xu et al. 2021). Additionally, diverse sizes of MPs were observed in the blood and urine of rodents in the current (Tsou et al. 2023) and other studies (Lee et al. 2022; Liang et al. 2021). Meanwhile, the MP signals have also been found in human feces (Schwabl et al. 2019; Zhang et al. 2021), placenta (Ragusa et al. 2021), and blood (Leslie et al. 2022), whereas some researchers are concerned with MP observations in human blood samples due to plastic contamination in common laboratory procedures (Jones et al. 2024; Kuhlman 2022).

MP-induced toxicity has been observed in diverse animals, including rodents. Several studies have demonstrated that MP induces adverse effects on the liver, intestine, kidney, spleen, and reproductive system of rodents (Amereh et al. 2020; Deng et al. 2017; Hou et al. 2021; Park et al. 2020; Wang et al. 2021). Some research revealed that MPs affect lipid homeostasis and glucose metabolism in mice by evaluating serum biochemical indicators and liver function biomarkers (Chen et al. 2022; Jin et al. 2019; Lu et al. 2018; Mu et al. 2022). Meanwhile, other studies have demonstrated that MPs induce oxidative stress, inflammation, and morphological changes, including the deposition of lipid droplets and hepatocyte damage in mice (Choi et al. 2021; Fan et al. 2022; Huang et al. 2022; Shi et al. 2022; Zheng et al. 2021). Although these studies illustrate MP-induced liver toxicity, some issues remain when discussing MP-induced hepatotoxicity. Most previous studies have examined the acute and subacute effects of MP exposure in mouse livers, whereas studies on the prolonged effects of MPs on the mouse liver are still limited. Additionally, these studies only partially examined MP-induced liver toxicity. MP-induced adverse effects, including perturbed metabolic function, liver inflammation, and liver pathology, were observed in diverse exposure models using different sizes, concentrations, and materials. Therefore, we examined the sub-chronic effects of MPs using a mouse model. The systematic toxicity assessment, including qualification and quantification of MPs in the liver and the status of oxidative stress, inflammation, biochemical homeostasis, and pathology, were monitored post-MP

exposure to draw a more comprehensive picture of MP-induced liver toxicity.

Materials and methods

Animals

The animal studies outlined below received approval from the Institutional Animal Care and Use Committee (IACUC) of the National Taiwan University's College of Medicine and College of Public Health, under permit number: 20201287. Female C57BL/6 mice, aged five weeks, were acquired from the Animal Center of the National Taiwan University School of Medicine. After a one-week acclimation period, the mice underwent MP exposure experiments. Throughout the experimental phase, the mice were maintained in a controlled environment with a 12-h light–dark cycle, $50 \pm 5\%$ humidity, and a temperature of 22 ± 2 °C. They were also provided with a standard diet (Laboratory Rodent Diet 5001, LabDiet) and sterile water.

Polystyrene microplastic exposure

The Nile Red-labeled, functional group-free, spherical sub-microscale-sized (0.5 μm), and microscale-sized (5 μm) polystyrene-MPs were purchased from SpheroTech (Lake Forest, USA). The characteristics of MP stock solutions were evaluated using an Attune NxT flow cytometer (Thermo Fisher Scientific, Waltham, MA, USA), ultra-high-resolution field emission-scanning electron microscopy (FE-SEM) in low-vacuum mode (FEI, Oregon, USA), zeta potential analyzer (DKSH & Malvern Zetasizer Nano ZSP), and dynamic light scattering (DKSH & Malvern Zetasizer Nano ZSP). Post dilution and ultrasonication, the final MP exposure concentrations were prepared as 0.3 mg per exposure for each size.

A total of 36 mice were randomly divided into three groups: 0.5 μm exposure ($n = 12$), 5 μm exposure ($n = 12$), and control ($n = 12$) groups. Four mice were housed in each cage. The exposure groups were treated with MPs by oral gavage, whereas the control group received Milli-Q water as the same method. The exposure dose of each MP was 0.3 mg per treatment, which was close to the releasing concentration of tea bag (Hernandez et al. 2019). Each mouse received MP solution or Milli-Q water twice a week for 12 consecutive weeks. The exposure concentration in mice was 0.6 mg per week, which was close to previous research (Choi et al. 2021; Fan et al. 2022). During the exposure period, the body weight and water consumption of each mouse were monitored. Post anesthetization and euthanasia, we first collected mouse serum and dissected and weighed the livers. The liver weight index for each mouse was calculated by

determining the ratio of liver weight to total body weight. For histopathological examination, half of the liver specimens were preserved in a 10% w/v formaldehyde solution in aqueous phosphate buffer. The remaining liver samples were instantly frozen in liquid nitrogen and kept at -80°C for subsequent analysis.

Quantitation of microplastics in mouse livers

The measurement of MP concentration was carried out in accordance with the methodology described in previous research with certain adjustments (Tsou et al. 2023). The livers of three mouse groups ($n = 12/\text{group}$) were examined. Equal amounts of liver tissues (25 mg) were lysed by 0.5X radioimmunoprecipitation buffer (100 μL ; $\text{pH} = 7.4$) and then dissociated by gentleMACS™ Octo Dissociator with Heaters (Miltenyi Biotec, Germany). Post homogenization and centrifugation (4°C , 3000 rpm, 3 min), the supernatant of the mixed solution was collected, diluted, and then vortexed before flow cytometry analysis. This study used the Attune NxT flow cytometer (Thermo Fisher Scientific, Waltham, MA, USA) with the optimized parameters for MP detection in mouse liver. Detailed experimental conditions are summarized in the supplementary materials.

Serum biochemistry analysis

Biochemical indicators in mouse serum were measured using a Roche Cobas c111 analyzer (Meylan, Isère, France). We evaluated the levels of albumin (ALB), alanine transaminase (ALT), aspartate transaminase (AST), creatinine (sCr), blood urea nitrogen (BUN), fasting glucose (GLU), total cholesterol (TCH), triglycerides (TG), high-density lipoprotein (HDL), and low-density lipoprotein, (LDL), which are extensively used to evaluate liver and kidney function, including energy and lipid metabolism homeostasis in human and animal studies.

Oxidative stress and inflammation

The superoxide dismutase (SOD) levels were evaluated using a mouse super oxidase dismutase enzyme-linked immunosorbent assay (ELISA) Kit (CSB-E08556m; Cusabio, Houston, TX, USA), according to a standard protocol. The biomarkers of oxidative damage, including 8-oxo-2'-deoxyguanosine (8-OHdG), 8-nitroguanine (8- NO_2Gua), 8-iso-prostaglandin $\text{F}_{2\alpha}$ (8-Iso $\text{PF}_{2\alpha}$), and 4-hydroxy-2-nonenal-mercapturic acid (HNE-MA), were measured using high-performance liquid chromatography-electrospray ionization tandem mass spectrometry as previously described (Wu et al. 2016).

The evaluation of the four inflammation markers, namely interferon- γ (IFN γ), interleukin-1 β (IL-1 β),

interleukin-6 (IL-6), and tumor necrosis factor- α (TNF α), was conducted using a mouse cytokine panel 2 multiplex ELISA Kit (Boster Biological Technology, CA, USA) and an Epoch Microplate Spectrophotometer (BioTek Instruments, Winooski, VT, USA), following standard protocols (Tsou et al. 2023).

Histopathology

The liver morphological changes were assessed by hematoxylin and eosin (H&E) staining ($n = 12/\text{group}$), Oil Red O staining ($n = 6/\text{group}$), and Masson's trichrome staining ($n = 12/\text{group}$). One pathological section from each mouse liver was examined for morphological changes or quantification.

Statistical analysis

Values are presented as the mean along with the standard error of the mean ($\text{mean} \pm \text{SEM}$). The comparison between the MP exposure group and the control group was performed using SAS statistical software (version 9.4). Due to the small number of groups, the Wilcoxon rank-sum test was applied. A p-value of less than 0.05 was considered to indicate statistically significant differences between the groups.

Results

Characteristics of Nile Red-labeled polystyrene microplastics

The scanning electron microscope views of 0.5 and 5 μm MPs were depicted in Fig. 1a, b, respectively. Meanwhile, the red spots in the fluorescence microscope views indicate the Nile Red-labeled 0.5 μm (Fig. 1c) and 5 μm (Fig. 1d) MPs. Specifically, the particle size range of 0.5 and 5 μm MPs were 0.40–0.54 μm and 5.62–5.69 μm . These two MP particles had spherical shapes with smooth surfaces. The comprehensive characteristics of the MPs, such as the average hydrodynamic diameter, surface charge, and polymer dispersity index, are presented in Fig. 1e, f and Supplementary Table S1.

MP accumulation in mouse livers

In the fluorescence microscope views, the MP accumulation in the mouse livers was found in the 0.5 μm exposure group (Fig. 2b), but not in the 5 μm exposure (Fig. 2c) and control groups (Fig. 2a). Similar results were observed with flow cytometry. The higher MPs were observed in the 0.5 μm exposure group (151,207 particles/g) than those in

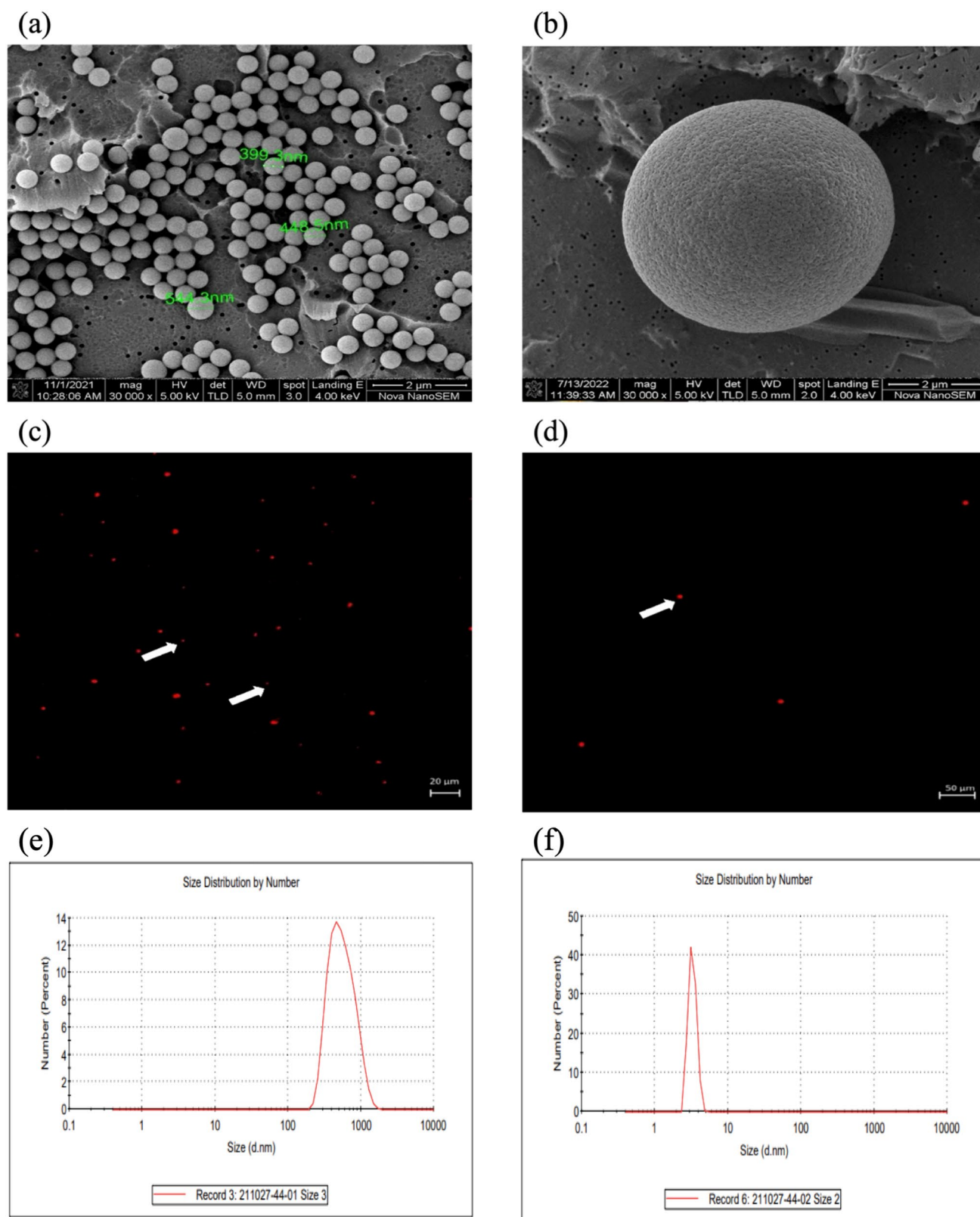


Fig. 1 Scanning electron microscope images of 0.5 µm (a) and 5 µm MPs (b); fluorescence microscope images of 0.5 µm (c) and 5 µm MPs (d)

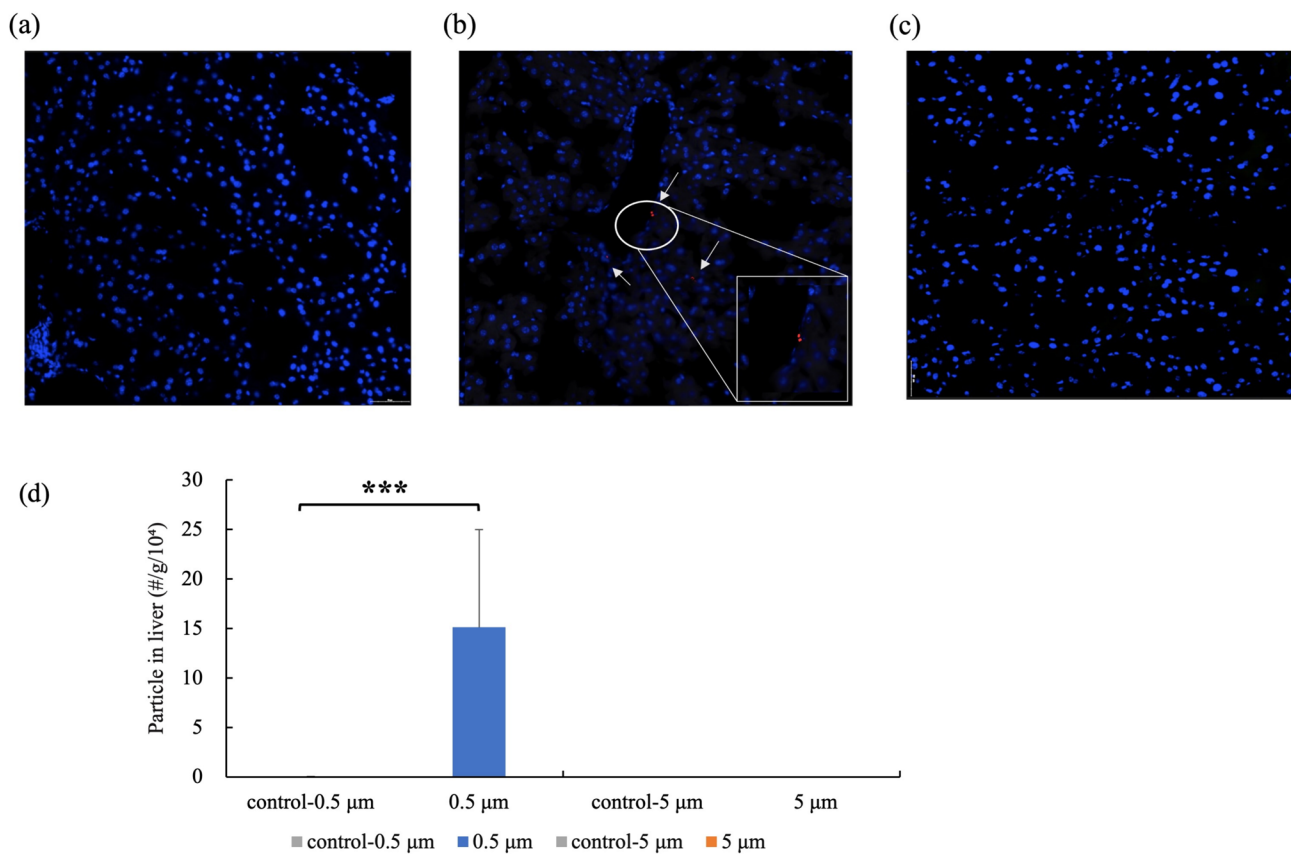


Fig. 2 Representative views of TissueFAXS in the livers of control (a), 0.5 μm (b), and 5 μm MPs (c) group mice. The detected MP numbers in mouse livers (d). The values are depicted as mean ± SEM. *** $p < 0.001$

the relative controls (222 particles/g) ($p < 0.001$), whereas no 5 μm-sized MP signals were detected in the mouse livers of both exposure and control groups.

Physiological condition

Throughout the duration of the exposure experiments, the total body weights of the mice across all three groups consistently rose (Fig. 3a). However, no noticeable weight changes were observed among the three groups. Meanwhile,

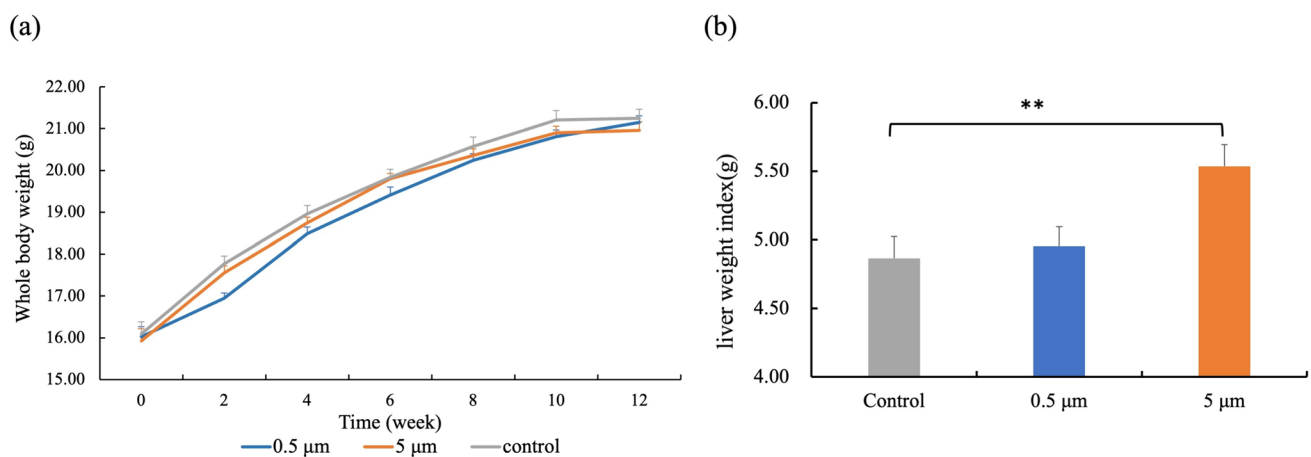


Fig. 3 Mean whole-body weight (a) and liver weight index (b) of mice among groups. The values are depicted as mean ± SEM. ** $p < 0.01$

the mouse liver weight index in the 5 μm exposure group was higher than those in the control one ($p < 0.01$), whereas no obvious differences in the 0.5 μm exposure groups compared to the control group were observed (Fig. 3b).

Histopathological changes of MPs in mouse livers

Haematoxylin and eosin staining revealed no obvious pathological changes in the mouse liver post-MP exposure. The histopathological changes observed in the Oil Red O staining test are depicted in Fig. 4a–f. More lipid droplets (red spots) were found in the mouse livers of the 0.5 μm exposure group (Fig. 4d, g) compared to the control group (Fig. 4b, g) ($p < 0.01$). However, this phenomenon was undetected in the 5 μm exposure group (Fig. 4f, g). Focal fibrosis in the liver was observed in three mice out of 12 in the 0.5 μm exposure group (Fig. 5b–d, f, g) by Masson's trichrome staining, whereas no obvious fibrosis was observed in mice of control (Fig. 5a, e) and 5 μm exposure group (data not shown).

Perturbed homeostasis of MPs in mouse serum

The homeostatic status was monitored using serum biochemical analyses. In the lipid metabolism, the increased HDL level in the 5 μm exposure group ($p < 0.05$) and increased TG level in the 0.5 μm exposure group ($p < 0.01$) were observed than those in control groups (Fig. 6a). Regarding

carbohydrate metabolism, increased glucose levels were found in both sized MP exposure groups ($p_{0.5 \mu\text{m}} < 0.01$; $p_{5 \mu\text{m}} < 0.05$) (Fig. 6b). In the liver function, the increased ALT ($p < 0.01$) and AST level ($p < 0.05$) in the 0.5 μm exposure group as well as a decreased AST level in the 5 μm exposure group ($p < 0.05$) were observed compared to the controls (Fig. 6c), whereas no obvious differences were found in the kidney-related biomarkers (Fig. 6d).

Status of oxidative stress and inflammation in mouse livers

The levels of the SOD proteins 8-OHdG, 8-NO₂Gua, HNE-MA, and 8-IsoPF_{2 α} in the mouse livers among the groups are depicted in Fig. 7. Among these oxidative stress biomarkers, higher levels of SOD proteins ($p < 0.05$) and HNE-MA were observed in the 0.5 μm exposure group ($p < 0.05$) than those in the control one, whereas a lower level of HNE-MA in the 5 μm exposure group ($p < 0.05$) compared to the control group were also observed. At the same time, the concentrations of IL-1 β , IL-6, IFN- γ , and TNF- α were assessed to determine the inflammatory condition of the mouse livers (Fig. 8). The level of IL-6 was significantly higher in the livers of mice in the 0.5 μm exposure group ($p < 0.01$) than in the livers of mice in the control group, whereas no obvious differences were found in the 5 μm exposure group compared to the control group.

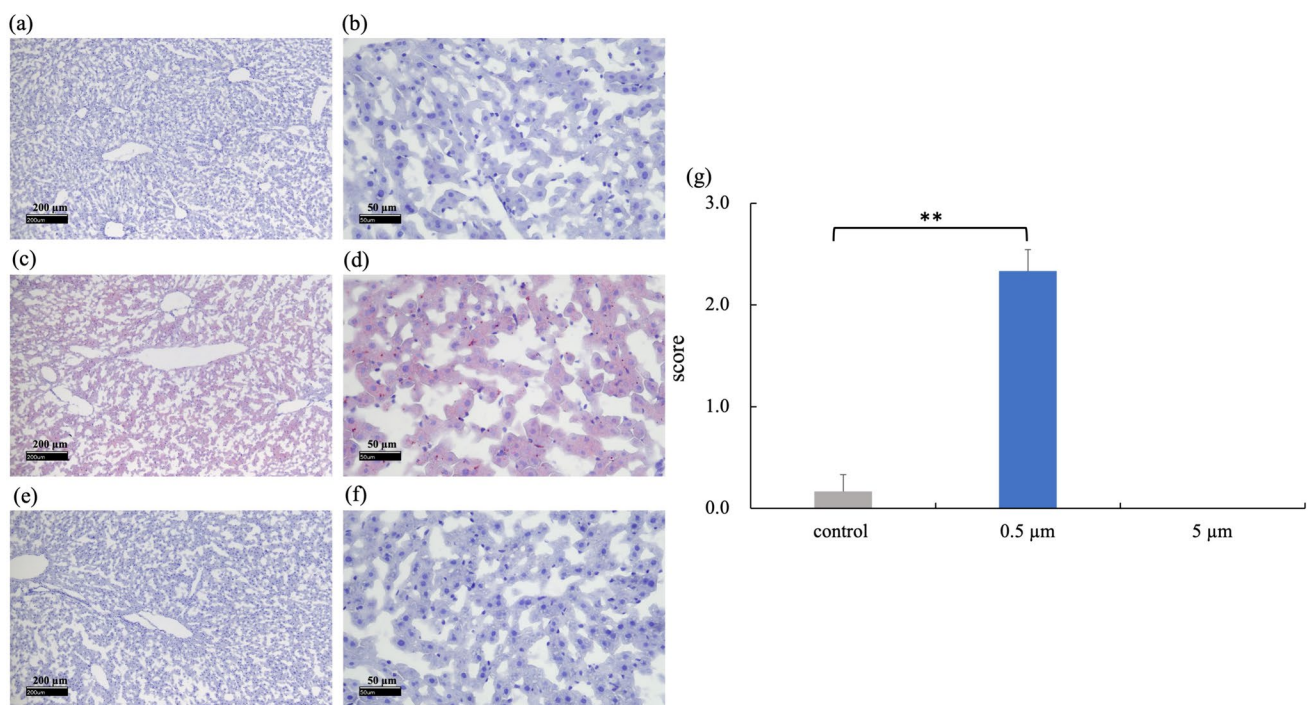


Fig. 4 Images of Oil red O staining test in mouse livers of control (a, b), 0.5 μm (c, d), and 5 μm MPs (e, f) group. The values are depicted as mean $\pm$ SEM. ** $p < 0.01$

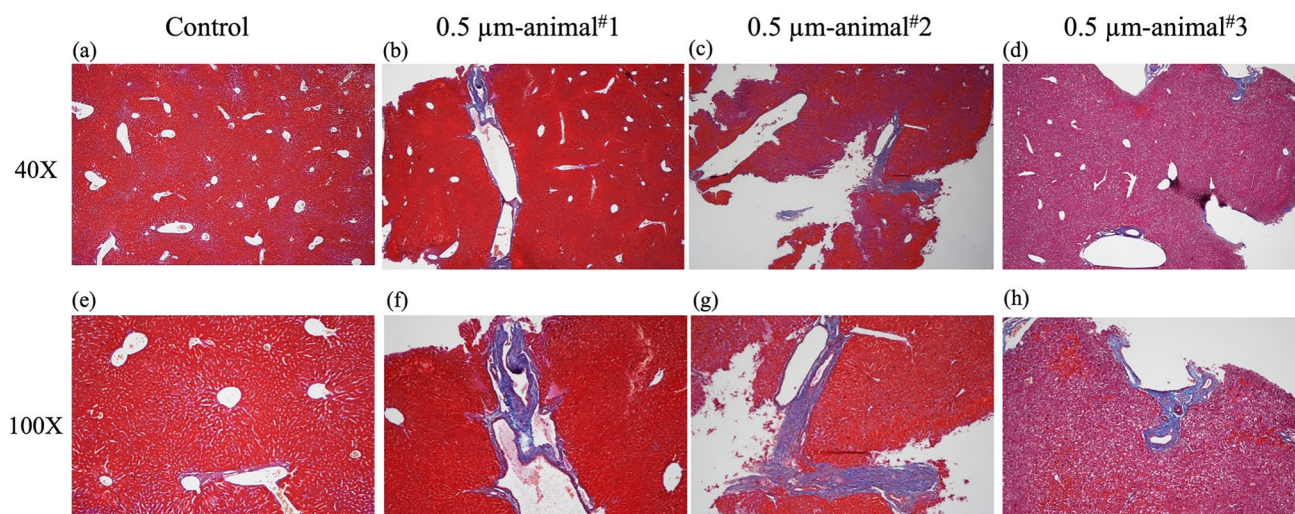


Fig. 5 Focal fibrosis in the liver is found in three mice out of 12 in the 0.5 μm exposure group (**b–d; f–h**) by Masson's trichrome staining, while no obvious fibrosis is observed in the control mice (**a, e**)

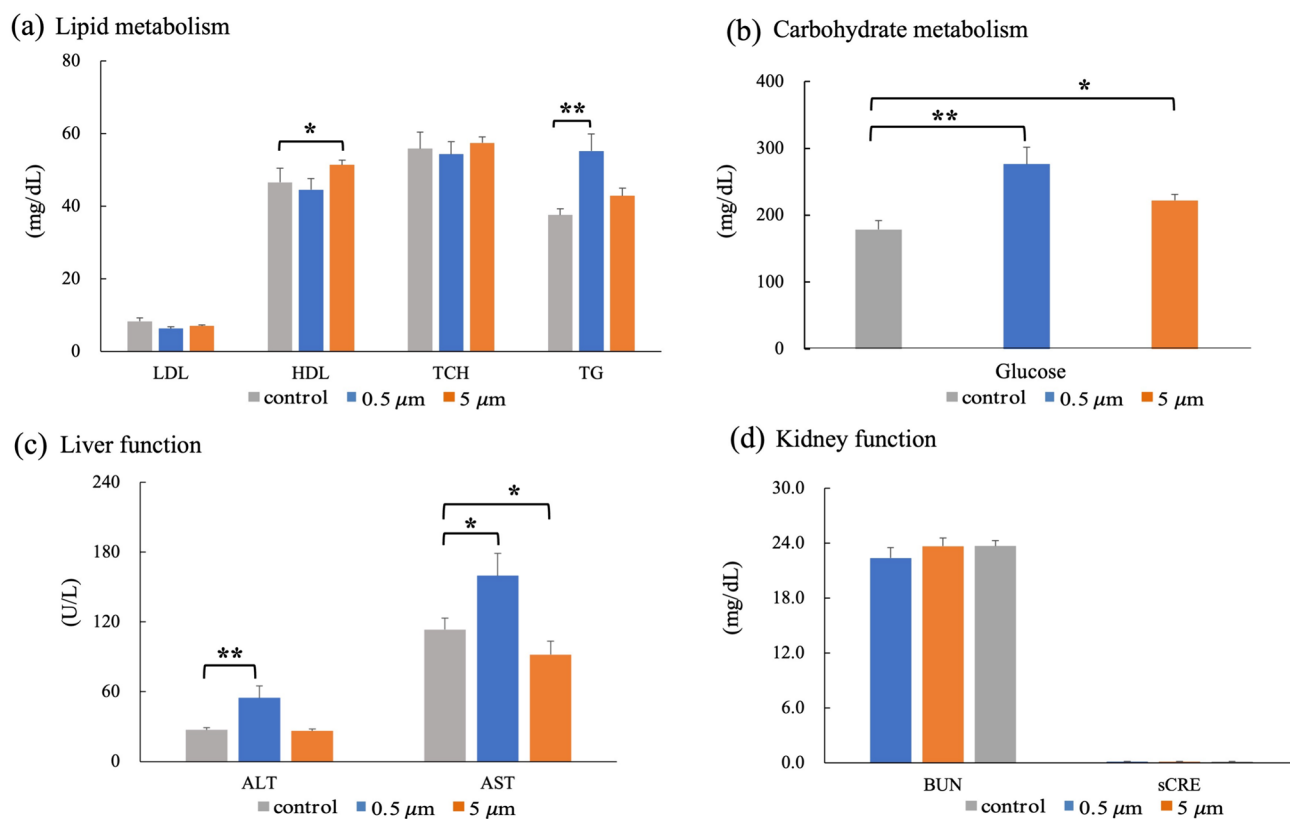


Fig. 6 The mouse serum biochemistry results in lipid metabolism (**a**), carbohydrate metabolism (**b**), liver function (**c**), and kidney (**d**) function. The values are depicted as mean $\pm$ SEM. * $p < 0.05$, ** $p < 0.01$

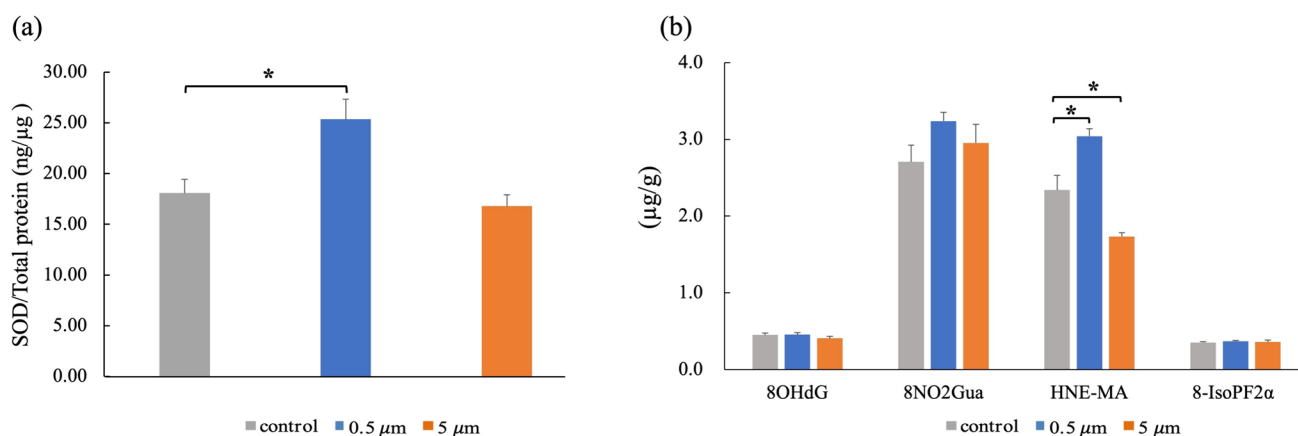
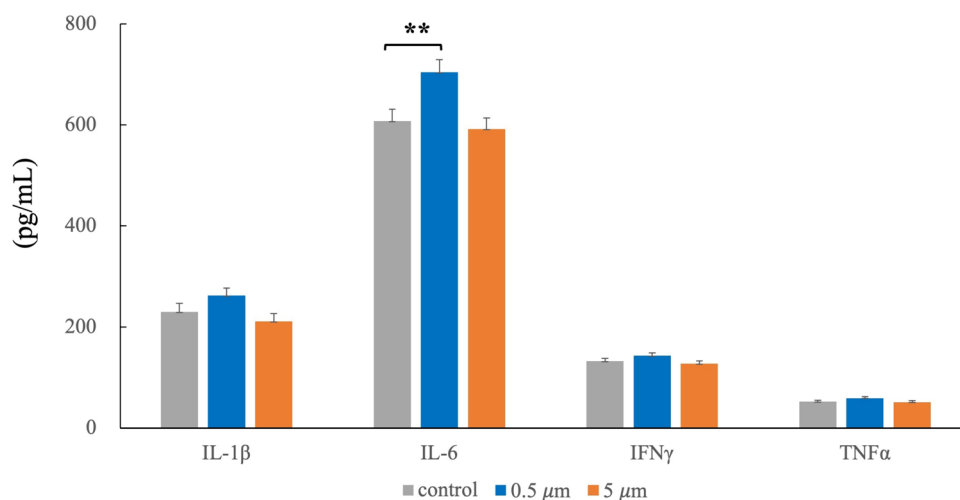


Fig. 7 Levels of SOD (a) and 8-OHdG, 8-NO₂Gua, HNE-MA, and 8-IsoPF_{2 α} (b) in the mouse livers among groups. The values are depicted as mean $\pm$ SEM. * $p < 0.05$

Fig. 8 Levels of IL-1 β , IL-6, IFN γ , and TNF- α in the mouse livers among groups. The values are depicted as mean $\pm$ SEM. ** $p < 0.01$



Discussion

In this study, we systematically evaluated the sub-chronic effects of oral gavage exposure of the mouse liver to MPs of diverse sizes. Our results demonstrated that submicron-sized MPs accumulated in the mouse liver. Additionally, lipid droplet accumulation, perturbed homeostasis of glucose and lipids, and damaged liver function, including increased oxidative stress and inflammation, were observed in the 0.5 μm MP exposure compared to those in control ones. Meanwhile, there was no MP accumulation, and limited cellular responses, such as increased glucose levels, HDL levels, and liver weight index, were observed in mice exposed to 5 μm MP. More detailed MP-mediated effects on the mouse liver are discussed in the following sections.

MP accumulation in the mouse livers was found in the 0.5 μm exposure group, but not in the 5 μm exposure group.

Numerous studies have detected submicron- and nano-sized MPs in mouse livers post-MP exposure (Choi et al. 2021; Fan et al. 2022; Liang et al. 2021; Shen et al. 2022; Tsou et al. 2023). However, micron-sized MP observations in the mouse liver have varied in diverse studies. Several research findings have shown the presence of micron-sized MPs in the livers of mice (Deng et al. 2017; Huang et al. 2022), but not in other studies (Stock et al. 2019), including the current study. Micron-sized MPs are more easily found in mouse livers than larger ones (Deng et al. 2017; Huang et al. 2022). These observations suggest that smaller MPs may have a higher potency to accumulate in the liver than micron-sized ones did.

In this study, we observed an increased liver weight index in the 5 μm exposure group, whereas obvious changes in the relevant indicators were not found. These phenomena imply that our 5 μm MP treatment induced a limited physiological perturbation in mice, although the

reasons for the increased liver weight were unclear. Previous studies also revealed the different responses of liver weight index. Similar to our results, the increased liver weight index was observed in mice in response to 5 μm MP treatment (Zhang et al. 2023), whereas the opposite changes were found in other 5 μm MP exposure experiments (Mu et al. 2022). These results indicated that liver weight index responses were inconsistent. More studies are needed to understand MP-induced liver weight index alterations. In addition, our results showed that 5 μm MPs were not accumulated in the mouse livers. This may imply that accumulated toxicants-induced biochemical and cytokine responses were not major reasons for the increased liver weight index. Other mechanisms, such as microbiota perturbation, also play a role in liver injuries. Previous studies indicated that MPs-induced hepatotoxicity through the gut microbiota (Zhang et al. 2023), which might be a possible reason.

This research observed elevated levels of oxidative stress markers, such as SOD and HNE-MA, in the liver tissues of mice exposed to MP, particularly noticeable in those exposed to the 0.5 μm size group. The SOD was one of the important enzymes for cells against free radicals (Yaribeygi et al. 2019). SOD protein overexpression suggests that liver cells require more anti-oxidative stress molecules to overcome the oxidative stress burden. Similar phenomena were also found in a mouse model with 0.5 μm polystyrene MP exposure (Choi et al. 2021), whereas decreased SOD levels were observed in previous studies owing to the differently sized and materialised MP exposures with diverse exposure durations (Chen et al. 2022; Fan et al. 2022; Mu et al. 2022; Shi et al. 2022). HNE-MA was one of the lipid peroxidation metabolites (Waits et al. 2020). HNE-MA accumulation in mouse livers indicated that oxidative stress existed in the MP exposure group. Increased levels of malondialdehyde (MDA), a lipid peroxidation biomarker, have been found in the mouse liver in response to exposure to diverse MP (Fan et al. 2022; Mu et al. 2022; Shen et al. 2022; Zheng et al. 2021). However, the decreased HNE-MA was observed in the 5 μm MP exposure group. Compared to the 0.5 μm MP exposure group, no 5 μm MP deposition was observed in this study, which means there are not enough external materials to increase the oxidative stress in mice. The adverse effects in mouse livers in the 5 μm MP exposure group may be mediated by other mechanisms, such as gut microbiota dysfunction (Zhang et al. 2023). This may result in a different status of oxidative stress. Our findings suggest that repeated exposure to 0.5 μm MPs causes MP accumulation in mouse livers, further triggers ongoing oxidative stress, and leads to additional harmful impacts on the liver of mice.

The increased levels of IL-6 in the 0.5 μm exposure group were similar to a previous study, which considered the same size and type of MPs with a shorter exposure period

(14 days) (Choi et al. 2021). Meanwhile, more inflammatory cytokines and their relative mRNA levels, including interleukin-1 α , IL-1 β , interleukin-18, interleukin-33, TNF- α , IFN- γ , and nuclear factor kappa B were elevated in the mouse livers and blood in response to MPs (Choi et al. 2021; Mu et al. 2022; Zhao et al. 2021; Zheng et al. 2021). As most previous studies have revealed MP-mediated liver inflammatory responses in acute-exposure models, our data demonstrated that long-term daily intake of MPs can still cause liver inflammation.

ALT and AST levels are common biomarkers used to evaluate liver function in clinical settings. Abnormal increases in these two markers indicate impaired liver function. Increased ALT and AST levels were found in this study and in several previous studies that conducted MP exposure for approximately 28–70 days (Huang et al. 2022; Mu et al. 2022; Shen et al. 2022; Zhao et al. 2021). However, this phenomenon was not observed in a mouse model with a 14-day exposure to MPs (Choi et al. 2021). These results suggest that prolonged exposure to MPs impairs liver function.

In our study, we observed that C57BL/6 mice exposed to microplastics had serum fasting glucose levels comparable to those observed in type 2 diabetes mellitus (T2DM). This suggests that MPs might not only alter glucose homeostasis but could also be a potential trigger for T2DM onset. Although our study did not directly evaluate insulin concentrations, hyperglycemia manifestation provides essential pathophysiological insights. The increased glucose levels, along with the observation that MP-exposed mice did not gain weight (indicating no obesity), suggested that the metabolic disruptions caused by MPs might have progressed beyond the early stages of insulin resistance (IR), typically marked by hyperinsulinemia. During the incipient stages of T2DM, there is augmented insulin production is a compensatory mechanism against IR. However, as the disease advances and the functional capacity of pancreatic β -cells diminishes (β -cell dysfunction), the compensatory insulin output becomes compromised (β -cells failure), resulting in pronounced hyperglycemia. Hence, our data infers that the MPs-exposed mice might have progressed beyond the hyperinsulinemic phase (pre-diabetes) and advanced to overt β -cell dysfunction/failure and its consequential hyperglycemia (Eizirik et al. 2020; Kahn 2003; Wysham and Shubrook 2020).

Our results, combined with those of previous studies, enrich our current understanding of the toxicological effects of MPs. Previous studies have demonstrated that MP exposure can elevate fasting blood glucose levels, insulin levels, and systemic IR, as indicated by metrics, including oral glucose tolerance test (OGTT), insulin tolerance test (ITT), and homeostasis model assessment of insulin resistance (HOMA-IR) index (Fan et al. 2022; Huang et al. 2022). However, detailed histological studies, such as those

by Huang et al. have underscored the effects of MP on the pancreas. Their data revealed that while control mice exhibited a normal pancreatic structure, MP-exposed mice had irregularly shaped and scattered islets with significant atrophy. This supports our theory that the adverse effects of MPs might go beyond the initial stages of IR, potentially reaching the advanced stages of T2DM. Moreover, the traditionally used markers for systemic IR may not be the best indicators for subjects already displaying high glucose levels (Kang et al. 2005; Katsuki et al. 2002).

Additionally, our investigation underscores the detrimental effects of MPs on lipid metabolism, which are marked by altered TGs and HDL levels following MP exposure. These lipid alterations, especially the perturbed TG levels, align with the typical lipid signatures seen in T2DM patients. The changes we noticed in glucose and lipid functions mirror the typical patterns observed in T2DM and its related lipid imbalances. Coupled with our primary findings regarding possible liver damage and the accumulation of MPs, these findings could intensify this complex metabolic relationship. If MPs boost oxidative stress and inflammation in the liver, they can exert additional strain on the ability of the liver to manage lipids. This may indirectly lead to the lipid irregularities commonly observed in T2DM. This prompted further investigation into whether MP exposure can accentuate lipid imbalances in T2DM and non-alcoholic fatty liver disease (NAFLD) populations (Matsuzaka and Shimano 2020).

NAFLD is increasingly becoming a primary contributor to chronic liver damage, although its mechanism of progression remains obscure. Although pathophysiological research indicates significant dietary and environmental influences on NAFLD development, the exact pathways remain obscure. Recent studies have suggested a potential link between MP exposure and NAFLD onset (Auguet et al. 2022; Zhou et al. 2023). In a study by Okamura et al., metabolic disturbances indicative of T2DM and NAFLD, such as impaired glucose tolerance and increased liver fat accumulation, were observed in C57BL/6 mice fed a high-fat diet (HFD). Interestingly, these disturbances were absent in mice fed a normal diet (ND). This finding highlights the potential synergistic effects of a contemporary lipid-rich diet and MPs in promoting metabolic conditions, such as T2DM and NAFLD (Okamura et al. 2023). Based on these findings, we identified parallel metabolic disturbances due to MPs in C57BL/6 mice on an ND, suggesting that the impact of MPs might be profound, even without the influence of an HFD. These observations underscore the urgency for comprehensive research on the influence of MP exposure on T2DM, dyslipidemia, and NAFLD.

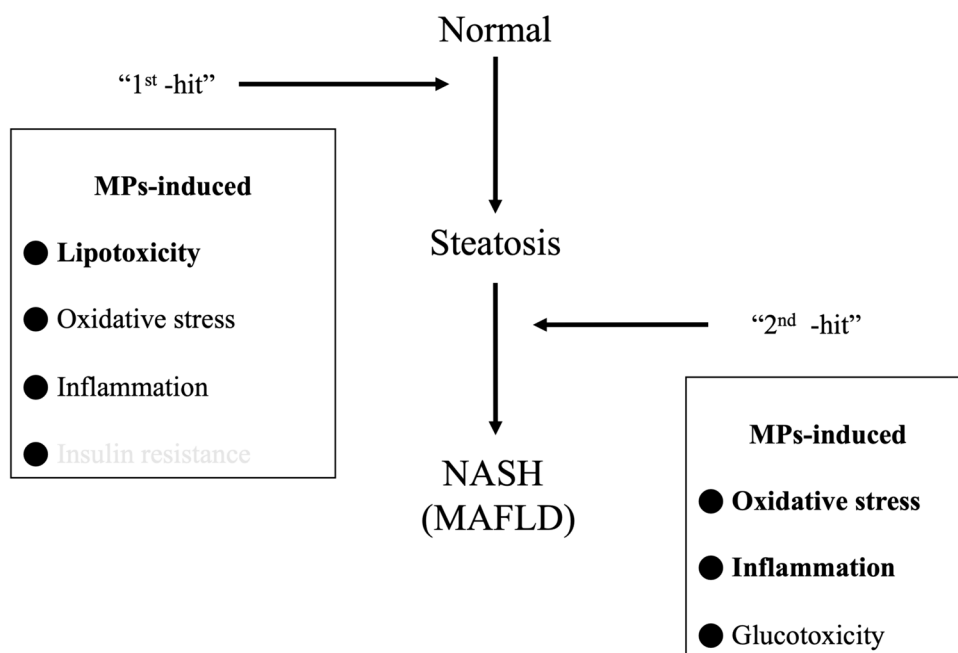
The intricate relationship among dyslipidemia, IR, NAFLD, and T2DM operates in a cyclical pattern. NAFLD progresses from steatosis to a more severe state of non-alcoholic steatohepatitis (NASH), potentially leading to cirrhosis and

hepatocellular carcinoma. Toxicant-associated fatty liver disease (TAFLD) is a pathological entity that shares characteristics with alcoholic liver disease and NAFLD. The more serious form of this condition, known as toxicant-associated steatohepatitis (TASH), is marked by the presence of hepatic steatosis, inflammation, and fibrosis. Notably, individuals with TASH are exposed to industrial chemicals, such as vinyl chloride (VC), without conventional risk factors such as alcohol intake or obesity. This progression is conceptualized by the "two-hit" or "multiple-hit" model. By integrating this model with our findings, we proposed that MPs serve as primary catalysts that trigger lipotoxicity, intrahepatic oxidative stress, and inflammation, leading to insulin resistance (the first hit). This is followed by the glucotoxicity pathway, intensified by β -cell dysfunction (hyperglycemia), marking the "second hit" (Fig. 9).

The experimental observations and associated pathological findings from this study may have implications for a range of hepatic disorders, including NASH, NAFLD, Metabolic-associated Fatty Liver Disease (MAFLD), TAFLD, and TASH. NAFLD may progress from steatosis and steatohepatitis to fibrosis and cirrhosis. In our study, lipid accumulation in the liver was evident, and histopathological analysis revealed focal fibrotic changes in certain animals subjected to MPs, which demonstrated that 0.5 μ m MPs could induce steatosis and steatohepatitis. This phenomenon can be exacerbated and cause more advanced liver damage, such as liver fibrosis, through the elongation of MP exposure time or increasing MP exposure doses. Focal fibrosis was found in some MP-treated mice, suggesting that subchronic exposure to MPs has the potential to perturb the homeostasis of liver function and further induce more severe liver injuries. Determining whether these changes strictly adhere to the clinical definitions of NAFLD or MAFLD should be approached with prudence (Brunt 2010; Eslam et al. 2020). Similarly, confirmation of whether focal fibrosis aligns with the criteria requires further investigation. Comprehensive investigations of the intrahepatic, cellular, and vascular responses are essential to deepen our understanding of the impact of MPs on liver disease.

Previous studies including the current study have revealed that MP-mediated liver lipid dysfunction occurs. (Huang et al. 2022; Lu et al. 2018; Zheng et al. 2021). However, current data only elucidate the effects of limited lipid species, especially TG. Additionally, the current research methods can only measure the total amount of TG levels, while the alterations in diverse TG and other lipid species have not been discussed. Therefore, a more comprehensive lipid profiling tool, such as lipidomics, is required to illustrate the MP-mediated lipid composition to further elucidate liver lipid dysfunction and diseases. Meanwhile, this study only examined polystyrene MP-induced liver toxicity, which can partially explain the MP effects on humans. Further studies

Fig. 9 Proposed mechanism of liver injury due to sub-chronic MP exposure. This diagram illustrates the potential progression of liver injury following exposure to microplastics. From MP-induced lipotoxicity and oxidative stress to the onset of insulin resistance, it further emphasizes the glucotoxicity pathway and β -cell dysfunction. The figure also highlights the absence of HFD, emphasizing the significant roles of sub-chronic MP exposure, lipotoxicity, and glucotoxicity in this pathological progression. This progression may be relevant to diverse forms of liver diseases, including non-alcoholic steatohepatitis (NASH) and metabolic-associated fatty liver disease (MAFLD)



NASH: non-alcoholic steatohepatitis

MAFLD: Metabolic-associated fatty liver disease

focusing on other MP materials can provide more evidence to elucidate the MP-mediated effects more clearly.

Conclusions

In conclusion, our results suggest that sub-chronic exposure to MPs induces MP accumulation in mouse livers and subsequently causes several liver dysfunctions, including increased oxidative stress, inflammation, impaired liver metabolic function on glucose and lipid metabolism, and poor performance in the clinical index of AST and ALT and abnormal lipid deposition in liver tissues simultaneously, which may suggest NASH-like toxicity. Additionally, our study revealed that smaller MPs induced greater effects in the mouse liver than larger ones did. Future studies should evaluate the effects of MP exposure duration on smaller MPs to better understand MP-mediated hepatotoxicity.

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Data availability Data will be made available on request from the corresponding author.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

- Akoueson F, Sheldon LM, Danopoulos E et al (2020) A preliminary analysis of microplastics in edible versus non-edible tissues from seafood samples. *Environ Pollut* 263:114452. <https://doi.org/10.1016/j.envpol.2020.114452>
- Allen S, Allen D, Phoenix VR et al (2019) Atmospheric transport and deposition of microplastics in a remote mountain catchment. *Nat Geosci* 12(5):339–344. <https://doi.org/10.1038/s41561-019-0335-5>
- Amereh F, Babaei M, Eslami A, Fazelpour S, Rafiee M (2020) The emerging risk of exposure to nano(micro)plastics on endocrine disturbance and reproductive toxicity: from a hypothetical scenario to a global public health challenge. *Environ Pollut* 261:114158. <https://doi.org/10.1016/j.envpol.2020.114158>
- Auguet T, Bertran L, Barrientos-Riosalido A et al (2022) are ingested or inhaled microplastics involved in nonalcoholic fatty liver disease? *Int J Environ Res Public Health*. <https://doi.org/10.3390/ijerph192013495>
- Avio CG, Pittura L, d'Errico G et al (2020) Distribution and characterization of microplastic particles and textile microfibers in adriatic food webs: general insights for biomonitoring strategies. *Environ Pollut* 258:113766. <https://doi.org/10.1016/j.envpol.2019.113766>
- Brunt EM (2010) Pathology of nonalcoholic fatty liver disease. *Nat Rev Gastroenterol Hepatol* 7(4):195–203. <https://doi.org/10.1038/nrgastro.2010.21>

- Chen X, Zhuang J, Chen Q, Xu L, Yue X, Qiao D (2022) Chronic exposure to polyvinyl chloride microplastics induces liver injury and gut microbiota dysbiosis based on the integration of liver transcriptome profiles and full-length 16S rRNA sequencing data. *Sci Total Environ* 839:155984. <https://doi.org/10.1016/j.scitotenv.2022.155984>
- Choi YJ, Park JW, Lim Y, Seo S, Hwang DY (2021) In vivo impact assessment of orally administered polystyrene nanoplastics: bio-distribution, toxicity, and inflammatory response in mice. *Nanotoxicology* 15(9):1180–1198. <https://doi.org/10.1080/17435390.2021.1996650>
- Deng Y, Zhang Y, Lemos B, Ren H (2017) Tissue accumulation of microplastics in mice and biomarker responses suggest widespread health risks of exposure. *Sci Rep* 7(1):46687. <https://doi.org/10.1038/srep46687>
- Ding Y, Zhang R, Li B et al (2021) Tissue distribution of polystyrene nanoplastics in mice and their entry, transport, and cytotoxicity to GES-1 cells. *Environ Pollut* 280:116974. <https://doi.org/10.1016/j.envpol.2021.116974>
- Du F, Cai H, Zhang Q, Chen Q, Shi H (2020) Microplastics in take-out food containers. *J Hazard Mater* 399:122969. <https://doi.org/10.1016/j.jhazmat.2020.122969>
- Eizirik DL, Pasquali L, Cnop M (2020) Pancreatic β -cells in type 1 and type 2 diabetes mellitus: different pathways to failure. *Nat Rev Endocrinol* 16(7):349–362. <https://doi.org/10.1038/s41574-020-0355-7>
- Eslam M, Newsome PN, Sarin SK et al (2020) A new definition for metabolic dysfunction-associated fatty liver disease: an international expert consensus statement. *J Hepatol* 73(1):202–209. <https://doi.org/10.1016/j.jhep.2020.03.039>
- Evangelidou N, Grythe H, Klimont Z et al (2020) Atmospheric transport is a major pathway of microplastics to remote regions. *Nat Commun* 11(1):3381. <https://doi.org/10.1038/s41467-020-17201-9>
- Fan X, Wei X, Hu H et al (2022) Effects of oral administration of polystyrene nanoplastics on plasma glucose metabolism in mice. *Chemosphere* 288(Pt 3):132607. <https://doi.org/10.1016/j.chemosphere.2021.132607>
- Fournier SB, D'Errico JN, Adler DS et al (2020) Nanopolystyrene translocation and fetal deposition after acute lung exposure during late-stage pregnancy. *Part Fibre Toxicol* 17(1):55. <https://doi.org/10.1186/s12989-020-00385-9>
- Hernandez LM, Xu EG, Larsson HCE, Tahara R, Maisuria VB, Tufenkji N (2019) Plastic teabags release billions of microparticles and nanoparticles into tea. *Environ Sci Technol* 53(21):12300–12310. <https://doi.org/10.1021/acs.est.9b02540>
- Horton AA, Walton A, Spurgeon DJ, Lahive E, Svendsen C (2017) Microplastics in freshwater and terrestrial environments: evaluating the current understanding to identify the knowledge gaps and future research priorities. *Sci Total Environ* 586:127–141. <https://doi.org/10.1016/j.scitotenv.2017.01.190>
- Hou J, Lei Z, Cui L et al (2021) Polystyrene microplastics lead to pyroptosis and apoptosis of ovarian granulosa cells via NLRP3/Caspase-1 signaling pathway in rats. *Ecotoxicol Environ Saf* 212:112012. <https://doi.org/10.1016/j.ecoenv.2021.112012>
- Huang D, Zhang Y, Long J et al (2022) Polystyrene microplastic exposure induces insulin resistance in mice via dysbacteriosis and proinflammation. *Sci Total Environ* 838(Pt 1):155937. <https://doi.org/10.1016/j.scitotenv.2022.155937>
- Hurley R, Woodward J, Rothwell JJ (2018) Microplastic contamination of river beds significantly reduced by catchment-wide flooding. *Nat Geosci* 11(4):251–257. <https://doi.org/10.1038/s41561-018-0080-1>
- Jin Y, Lu L, Tu W, Luo T, Fu Z (2019) Impacts of polystyrene microplastic on the gut barrier, microbiota and metabolism of mice. *Sci Total Environ* 649:308–317. <https://doi.org/10.1016/j.scitotenv.2018.08.353>
- Jones NR, de Jersey AM, Lavers JL, Rodemann T, Rivers-Auty J (2024) Identifying laboratory sources of microplastic and nanoplastic contamination from the air, water, and consumables. *J Hazard Mater* 465:133276. <https://doi.org/10.1016/j.jhazmat.2023.133276>
- Kahn SE (2003) The relative contributions of insulin resistance and beta-cell dysfunction to the pathophysiology of type 2 diabetes. *Diabetologia* 46(1):3–19. <https://doi.org/10.1007/s00125-002-1009-0>
- Kang ES, Yun YS, Park SW et al (2005) Limitation of the validity of the homeostasis model assessment as an index of insulin resistance in Korea. *Metabolism* 54(2):206–211. <https://doi.org/10.1016/j.metabol.2004.08.014>
- Katsuki A, Sumida Y, Urakawa H et al (2002) Neither homeostasis model assessment nor quantitative insulin sensitivity check index can predict insulin resistance in elderly patients with poorly controlled type 2 diabetes mellitus. *J Clin Endocrinol Metab* 87(11):5332–5335. <https://doi.org/10.1210/jc.2002-020486>
- Kedzierski M, Lechat B, Sire O, Le Maguer G, Le Tilly V, Bruzaud S (2020) Microplastic contamination of packaged meat: occurrence and associated risks. *Food Packag Shelf Life* 24:100489. <https://doi.org/10.1016/j.fpsl.2020.100489>
- Kelly A, Lannuzel D, Rodemann T, Meiners KM, Auman HJ (2020) Microplastic contamination in east Antarctic sea ice. *Mar Pollut Bull* 154:111130. <https://doi.org/10.1016/j.marpolbul.2020.111130>
- Kim J-S, Lee H-J, Kim S-K, Kim H-J (2018) Global pattern of microplastics (MPs) in commercial food-grade salts: sea salt as an indicator of seawater MP pollution. *Environ Sci Technol* 52(21):12819–12828. <https://doi.org/10.1021/acs.est.8b04180>
- Kuhlman RL (2022) Letter to the editor, discovery and quantification of plastic particle pollution in human blood. *Environ Int* 167:107400. <https://doi.org/10.1016/j.envint.2022.107400>
- Lebreton L, Slat B, Ferrari F et al (2018) Evidence that the Great Pacific Garbage Patch is rapidly accumulating plastic. *Sci Rep* 8(1):4666. <https://doi.org/10.1038/s41598-018-22939-w>
- Lee S, Kang K-K, Sung S-E et al (2022) Toxicity study and quantitative evaluation of polyethylene microplastics in ICR mice. *Polymers*. <https://doi.org/10.3390/polym14030402>
- Leslie HA, van Velzen MJM, Brandsma SH, Vethaak AD, Garcia-Vallejo JJ, Lamoree MH (2022) Discovery and quantification of plastic particle pollution in human blood. *Environ Int* 163:107199. <https://doi.org/10.1016/j.envint.2022.107199>
- Li D, Shi Y, Yang L et al (2020a) Microplastic release from the degradation of polypropylene feeding bottles during infant formula preparation. *Nat Food* 1(11):746–754. <https://doi.org/10.1038/s43016-020-00171-y>
- Li Q, Feng Z, Zhang T, Ma C, Shi H (2020b) Microplastics in the commercial seaweed nori. *J Hazard Mater* 388:122060. <https://doi.org/10.1016/j.jhazmat.2020.122060>
- Liang B, Zhong Y, Huang Y et al (2021) Underestimated health risks: polystyrene micro- and nanoplastics jointly induce intestinal barrier dysfunction by ROS-mediated epithelial cell apoptosis. *Part Fibre Toxicol* 18(1):20. <https://doi.org/10.1186/s12989-021-00414-1>
- Lu L, Wan Z, Luo T, Fu Z, Jin Y (2018) Polystyrene microplastics induce gut microbiota dysbiosis and hepatic lipid metabolism disorder in mice. *Sci Total Environ* 631–632:449–458. <https://doi.org/10.1016/j.scitotenv.2018.03.051>
- Matsuzaka T, Shimano H (2020) New perspective on type 2 diabetes, dyslipidemia and non-alcoholic fatty liver disease. *J Diabetes Investig* 11(3):532–534. <https://doi.org/10.1111/jdi.13258>
- Mu Y, Sun J, Li Z et al (2022) Activation of pyroptosis and ferroptosis is involved in the hepatotoxicity induced by polystyrene microplastics in mice. *Chemosphere* 291:132944. <https://doi.org/10.1016/j.chemosphere.2021.132944>

- Okamura T, Hamaguchi M, Hasegawa Y et al (2023) Oral exposure to polystyrene microplastics of mice on a normal or high-fat diet and intestinal and metabolic outcomes. *Environ Health Perspect* 131(2):27006. <https://doi.org/10.1289/ehp11072>
- Oliveri Conti G, Ferrante M, Banni M et al (2020) Micro- and nano-plastics in edible fruit and vegetables. The first diet risks assessment for the general population. *Environ Res* 187:109677. <https://doi.org/10.1016/j.envres.2020.109677>
- Park EJ, Han JS, Park EJ et al (2020) Repeated-oral dose toxicity of polyethylene microplastics and the possible implications on reproduction and development of the next generation. *Toxicol Lett* 324:75–85. <https://doi.org/10.1016/j.toxlet.2020.01.008>
- Pohl F, Eggenhuisen JT, Kane IA, Clare MA (2020) Transport and burial of microplastics in deep-marine sediments by turbidity currents. *Environ Sci Technol* 54(7):4180–4189. <https://doi.org/10.1021/acs.est.9b07527>
- Prata JC, Paço A, Reis V et al (2020) Identification of microplastics in white wines capped with polyethylene stoppers using micro-Raman spectroscopy. *Food Chem* 331:127323. <https://doi.org/10.1016/j.foodchem.2020.127323>
- Qi R, Jones DL, Li Z, Liu Q, Yan C (2020) Behavior of microplastics and plastic film residues in the soil environment: a critical review. *Sci Total Environ* 703:134722. <https://doi.org/10.1016/j.scitotenv.2019.134722>
- Ragusa A, Svelato A, Santacroce C et al (2021) Plasticenta: first evidence of microplastics in human placenta. *Environ Int* 146:106274. <https://doi.org/10.1016/j.envint.2020.106274>
- Ranjan VP, Joseph A, Goel S (2021) Microplastics and other harmful substances released from disposable paper cups into hot water. *J Hazard Mater* 404:124118. <https://doi.org/10.1016/j.jhazmat.2020.124118>
- Revel M, Châtel A, Mouneyrac C (2018) Micro(nano)plastics: a threat to human health? *Curr Opin Environ Sci Health* 1:17–23. <https://doi.org/10.1016/j.coesh.2017.10.003>
- Schwabl P, Köppel S, Königshofer P et al (2019) Detection of various microplastics in human stool. *Ann Intern Med* 171(7):453–457. <https://doi.org/10.7326/M19-0618>
- Schymanski D, Goldbeck C, Humpf H-U, Fürst P (2018) Analysis of microplastics in water by micro-Raman spectroscopy: release of plastic particles from different packaging into mineral water. *Water Res* 129:154–162. <https://doi.org/10.1016/j.watres.2017.11.011>
- Shen R, Yang K, Cheng X et al (2022) Accumulation of polystyrene microplastics induces liver fibrosis by activating cGAS/STING pathway. *Environ Pollut* 300:118986. <https://doi.org/10.1016/j.envpol.2022.118986>
- Shi C, Han X, Guo W et al (2022) Disturbed gut-liver axis indicating oral exposure to polystyrene microplastic potentially increases the risk of insulin resistance. *Environ Int* 164:107273. <https://doi.org/10.1016/j.envint.2022.107273>
- Shruti VC, Pérez-Guevara F, Elizalde-Martínez I, Kutralam-Muniasamy G (2020) First study of its kind on the microplastic contamination of soft drinks, cold tea and energy drinks—future research and environmental considerations. *Sci Total Environ* 726:138580. <https://doi.org/10.1016/j.scitotenv.2020.138580>
- Stock V, Böhmert L, Lisicki E et al (2019) Uptake and effects of orally ingested polystyrene microplastic particles in vitro and in vivo. *Arch Toxicol* 93(7):1817–1833. <https://doi.org/10.1007/s00204-019-02478-7>
- Tsou TY, Lee SH, Kuo TH, Chien CC, Chen HC, Cheng TJ (2023) Distribution and toxicity of submicron plastic particles in mice. *Environ Toxicol Pharmacol* 97:104038. <https://doi.org/10.1016/j.etap.2022.104038>
- Waits A, Chen H-C, Kuo P-L et al (2020) Urinary phthalate metabolites are associated with biomarkers of DNA damage and lipid peroxidation in pregnant women—Tainan birth cohort study (TBCS). *Environ Res* 188:109863. <https://doi.org/10.1016/j.envres.2020.109863>
- Walczak AP, Hendriksen PJM, Woutersen RA et al (2015) Bioavailability and biodistribution of differently charged polystyrene nanoparticles upon oral exposure in rats. *J Nanopart Res* 17(5):231. <https://doi.org/10.1007/s11051-015-3029-y>
- Wang Y-L, Lee Y-H, Hsu Y-H et al (2021) The kidney-related effects of polystyrene microplastics on human kidney proximal tubular epithelial cells HK-2 and male C57BL/6 mice. *Environ Health Perspect* 129(5):057003. <https://doi.org/10.1289/EHP7612>
- Wu C, Chen ST, Peng KH, Cheng TJ, Wu KY (2016) Concurrent quantification of multiple biomarkers indicative of oxidative stress status using liquid chromatography-tandem mass spectrometry. *Anal Biochem* 512:26–35. <https://doi.org/10.1016/j.ab.2016.07.030>
- Wysham C, Shubrook J (2020) Beta-cell failure in type 2 diabetes: mechanisms, markers, and clinical implications. *Postgrad Med* 132(8):676–686. <https://doi.org/10.1080/00325481.2020.1771047>
- Xu D, Ma Y, Han X, Chen Y (2021) Systematic toxicity evaluation of polystyrene nanoplastics on mice and molecular mechanism investigation about their internalization into Caco-2 cells. *J Hazard Mater* 417:126092. <https://doi.org/10.1016/j.jhazmat.2021.126092>
- Yaribeygi H, Atkin SL, Sahebkar A (2019) A review of the molecular mechanisms of hyperglycemia-induced free radical generation leading to oxidative stress. *J Cell Physiol* 234(2):1300–1312. <https://doi.org/10.1002/jcp.27164>
- Zhang K, Yang J, Chen L et al (2023) Gut microbiota participates in polystyrene microplastics-induced hepatic injuries by modulating the gut-liver axis. *ACS Nano* 17(15):15125–15145. <https://doi.org/10.1021/acsnano.3c04449>
- Zhang N, Li YB, He HR, Zhang JF, Ma GS (2021) You are what you eat: Microplastics in the feces of young men living in Beijing. *Sci Total Environ* 767:144345. <https://doi.org/10.1016/j.scitotenv.2020.144345>
- Zhao L, Shi W, Hu F, Song X, Cheng Z, Zhou J (2021) Prolonged oral ingestion of microplastics induced inflammation in the liver tissues of C57BL/6J mice through polarization of macrophages and increased infiltration of natural killer cells. *Ecotoxicol Environ Saf* 227:112882. <https://doi.org/10.1016/j.ecoenv.2021.112882>
- Zheng H, Wang J, Wei X, Chang L, Liu S (2021) Proinflammatory properties and lipid disturbance of polystyrene microplastics in the livers of mice with acute colitis. *Sci Total Environ* 750:143085. <https://doi.org/10.1016/j.scitotenv.2020.143085>
- Zhou W, Shi W, Du X et al (2023) Assessment of nonalcoholic fatty liver disease symptoms and gut-liver axis status in zebrafish after exposure to polystyrene microplastics and oxytetracycline, alone and in combination. *Environ Health Perspect* 131(4):47006. <https://doi.org/10.1289/ehp11600>

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